Unit 2 Progress Check: FRQ Part A

This question involves processing a String that represents a sequence of beads.

The String variable beadSequence represents a bead sequence. It contains a sequence of letters in which each letter represents the color of a bead.

A bead sequence is considered acceptable if it adheres to the following rules. Otherwise, it is not considered acceptable.

- Rule 1: The number of beads in the sequence is between 5 and 10, inclusive.
- Rule 2: The sequence contains at least one blue bead (represented by "B") and at least one white bead (represented by "W").
- Rule 3: The sequence starts and ends with the same color.

You will create three boolean variables to help determine whether the sequence is acceptable, and then print a message indicating the result. The following table illustrates the intended behavior for different values of beadSequence.

Value of beadSequence	Meets Rule 1	Meets Rule 2	Meets Rule 3	Printed Output
"OBWO"	false	true	true	"Not acceptable"
"BWBGOGWBB"	true	true	true	"Acceptable"
"ROYGBIVW"	true	true	false	"Not acceptable"
"BOBOBOBO"	true	false	false	"Not acceptable"
"WBWRWBWRWBWR"	false	true	false	"Not acceptable"
"BWWWBBWWWB"	true	true	true	"Acceptable"

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1. Write a code segment that creates a boolean variable named lengthGood and assigns it true if beadSequence meets rule 1 and assigns it false otherwise.

Part A Point 1

Select a point value to view scoring criteria or to score the response.

To earn this point, responses must demonstrate the following:

- Finds the length of the bead sequence using appropriate `String` method call

Canonical Solution:

```
int len = beadSequence.length();
boolean lengthGood = (len >= 5 && len <= 10);
```

0	1 ✓

The response earns the point.

Responses will not earn the point if they

- make any incorrect call to a `String` method

Part A Point 2

Select a point value to view scoring criteria or to score the response.

To earn this point, responses must demonstrate the following:

- Assigns the appropriate `boolean` value to `lengthGood` based on the length of the bead sequence

Canonical Solution:

```
int len = beadSequence.length();
boolean lengthGood = (len >= 5 && len <= 10);
```

0	1 ✓

The response earns the point.

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Responses can still earn the point if they

- incorrectly determine the length of the bead sequence

Responses will not earn the point if they

- fail to compare against the length of `5` and the length of `10`

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2. Write a code segment that creates a boolean variable named correctColors and assigns it true if beadSequence meets rule 2 and assigns it false otherwise.

Part B Point 1

Select a point value to view scoring criteria or to score the response.

To earn this point, responses must demonstrate the following:

- Uses an appropriate `String` method to determine the bead sequence contains at least one `"B"` and at least one `"W"`

Canonical Solution:

```
boolean correctColors;
if ((beadSequence.indexOf("B") == -1) ||
    (beadSequence.indexOf("W") == -1))
{
   correctColors = false;
}
else
{
   correctColors = true;
}
```

0	1 ✓

The response earns the point.

Responses will not earn the point if they

- make any incorrect call to a `String` method
- fail to test for both `"B"` and `"W"`

Part B Point 2

Select a point value to view scoring criteria or to score the response.

To earn this point, responses must demonstrate the following:

- Assigns the appropriate `boolean` value to `correctColors` based on the values `"B"` and `"W"` both appearing in the bead sequence

Canonical Solution:

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```
boolean correctColors;
if ((beadSequence.indexOf("B") == -1) ||
    (beadSequence.indexOf("W") == -1))
{
   correctColors = false;
}
else
{
   correctColors = true;
}
```

0	✓ 1

The response earns the point.

Responses can still earn the point if they

- make any incorrect call to a `String` method
- fail to test for both `"B"` and `"W"`

Responses will not earn the point if they

- fail to check for letters appearing in the bead sequence

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3. Write a code segment that creates a boolean variable named startEndGood and assigns it true if beadSequence meets rule 3 and assigns it false otherwise.

Part C Point 1

Select a point value to view scoring criteria or to score the response.

To earn this point, responses must demonstrate the following:

- Uses an appropriate `String` method call to identify the first and last letters of the bead sequence

Canonical Solution:

```
String first = beadSequence.substring(0, 1);
String last = beadSequence.substring(len - 1);
boolean startEndGood = (first == last);
```

0	1 ✓

The response earns the point.

Responses will not earn the point if they

- make any incorrect call to a `String` method
- fail to consider both first and last letters

Part C Point 2

Select a point value to view scoring criteria or to score the response.

To earn this point, responses must demonstrate the following:

- Assigns the appropriate `boolean` value to `startEndGood` based on the identified first and last letters

Canonical Solution:

```
String first = beadSequence.substring(0, 1);
String last = beadSequence.substring(len - 1);
boolean startEndGood = (first == last);
```

0	1 ✓

The response earns the point.

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Responses can still earn the point if they

- make an incorrect call to a `String` method
- incorrectly identify first or last letter

Responses will not earn the point if they

- fail to consider both first and last letters

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4. Write a code segment that prints "Acceptable" if beadSequence adheres to all three rules and prints "Not acceptable" otherwise. You must use the variables lengthGood, correctColors, and startEndGood to receive full credit, regardless of what you wrote for parts A, B, and C.

Part D Point 1

Select a point value to view scoring criteria or to score the response.

To earn this point, responses must demonstrate the following:

- Determines if bead sequence is acceptable using the variables `lengthGood`, `correctColors`, and `startEndGood`

Canonical Solution:

```
if (lengthGood && correctColors && startEndGood)
{
   System.out.println("Acceptable");
}
else
{
   System.out.println("Not acceptable");
}
```

0	1 ✓

The response earns the point.

Responses will not earn the point if they

- fail to use `lengthGood`, `correctColors`, and `startEndGood` variables

Part D Point 2

Select a point value to view scoring criteria or to score the response.

To earn this point, responses must demonstrate the following:

- Prints message based on determination of whether bead sequence is acceptable

Canonical Solution:

```
if (lengthGood && correctColors && startEndGood)
{
```

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```
   System.out.println("Acceptable");
}
else
{
   System.out.println("Not acceptable");
}
```

0	1 ✓

The response earns the point.

Responses can still earn the point if they

- incorrectly determine if the bead sequence is acceptable

Responses will not earn the point if they

- fail to print the message